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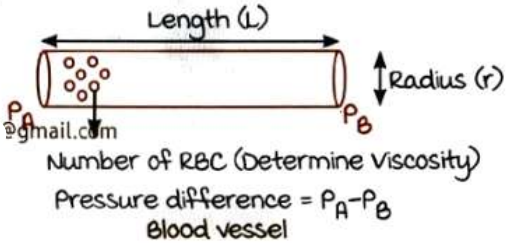
HEMODYNAMICS

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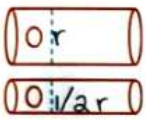
Poiseuille-Hagen Law

00:02:07

$$\text{Blood flow} \propto \frac{(P_A - P_B) \times (\text{Radius})^4}{\text{Length} \times \text{viscosity}}$$



1. Blood flow $\propto r^4$

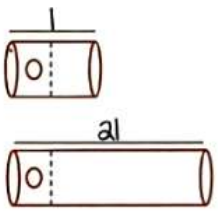


Normal blood vessel

Renal artery stenosis

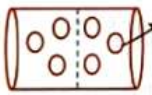
Blood flow reduced by 16 times

2. Blood flow $\propto 1/L$



Blood flow reduced to 1/2

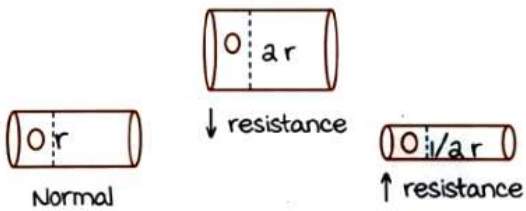
3. Blood flow $\propto$ 1/viscosity



↑ RBC number (↑ viscosity)

Blood flow in ↓ hyperviscosity

4. Fourth power law: Resistance $\propto 1/\text{radius}^4$



↓ resistance

Normal

↑ resistance

5. Blood flow $\propto$ 1/area

Blood Pressure

00:11:50

The lateral pressure exerted by the blood on a vessel wall.

PRINCIPLE:

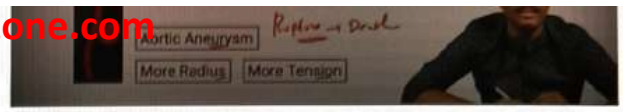
Bernoulli's principle: Potential energy + Kinetic energy = Constant.

(Blood pressure) (Blood flow)

- Hence BP + Blood flow = Constant.



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MEASUREMENT :

SI unit : Pascal.

Preferred unit : mm Hg

- Systolic blood pressure = 120 mm Hg
- Diastolic blood pressure = 80 mm Hg

Formulas :

- Pulse pressure : SBP-DBP.
- mean arterial pressure (MAP) = DBP + 1/3 pulse pressure = 93-100 mm Hg.

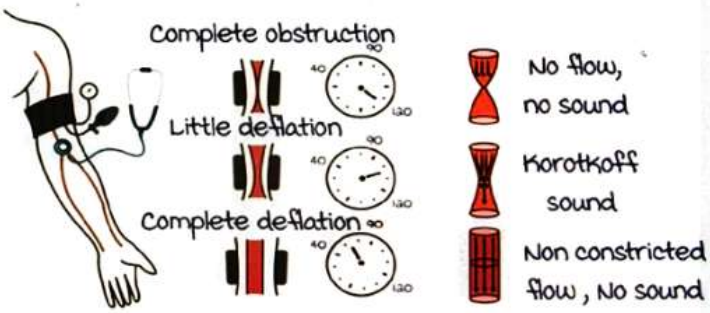
Modes of measurement :

Direct	Indirect
• Invasive. • measure intraarterial BP. • used in animals (Research purpose).	• Non invasive. • Underestimates SBP by 10 mm Hg. • used in clinical practice.)gmail.com


sphygmomanometer

Indirect method :

- Korotkoff's sounds :
  - Phase 1 : Faint tapping sound (SBP).
  - Phase 2 : Murmur like sound.
  - Phase 3 : Crisp & clear sound.
  - Phase 4 : Muffled sound.
  - Phase 5 : Sound disppears (DBP).

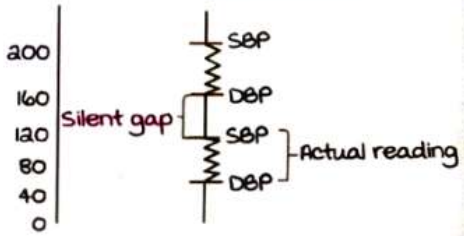


- False BP recordings :
  - Small size cuff used.
  - Obesity.
  - Thick & calcified vessel.

- Length : Cover 80% of arm circumference.
- Width : Cover 40% of arm circumference.


Riva Roci cuff

- Auscultatory gap :
  - Significance : DBP overestimated.



Feedback

Direct method :

Blood pressure in heart chambers :

Heart chambers	SBP (mm Hg)	DBP (mm Hg)
Right atrium	4	0
Right ventricle	25	5
Left atrium	12	4
Left ventricle	140	4

Pulmonary capillary wedge pressure (PCWP) :

- Measure left atrial pressure : 4–12 mm Hg (Average 10 mm Hg).

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Other Parameters Determining Blood Flow

00:26:50

1. REYNOLD'S NUMBER :



→ Laminar flow (Silent)

→ Turbulent flow (Noise present)

- Seen in aorta & pulmonary artery.
- Basis of murmurs.

- Reynold's number α $\frac{\text{Density of blood} \times \text{Diameter} \times \text{Velocity of flow}}{\text{Viscosity}}$
- Use : Helps to predict type of blood flow.
- Interpretation :

<2000	2000–3000	>3000
Laminar flow	Transitional flow	Turbulent flow

- Clinical significance : In pregnancy – flow murmurs heard (D/t anemia, ↓ RBC : ↓ viscosity → ↑ Reynold's number.

2. LAPLACE'S LAW

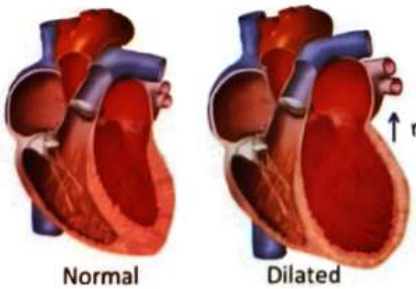
Pressure (P) = 2 x Tension (T)/ Radius (r).

Clinical significance :

Tension α Radius



Small radius → Thin wall → (D/t tension) → Doesn't rupture



↑ radius : ↑ tension

Normal Dilated

Dilated cardiomyopathy → (D/t tension) → Heart failure



↑ radius

Aortic aneurysm → (D/t tension) → Rupture → Death

Feedback